Lecture 10

Taylor and Maclaurin Polynomials and Series

Taylor Polynomial: If f can be differentiated n times at x_0, then we define the nth Taylor polynomial for f about $x = x_0$ to be

$$\begin{aligned} P_n(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \\ &= \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \quad \text{[Summation Form]} \end{aligned}$$

Here,

$$\begin{aligned} P_0(x) &= f(x_0) \\ P_1(x) &= f(x_0) + f'(x_0)(x - x_0) \\ P_2(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 \\ P_3(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 \\ &\vdots \end{aligned}$$

Taylor Polynomial of a function $f(x)$ gives a polynomial approximation of $f(x)$ near a point x_0. It gives the best possible polynomial that behaves like $f(x)$ near x_0.

Taylor Series: If f has derivatives of all orders at x_0, then we call the series

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \cdots \\ &\quad + \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \cdots \end{aligned}$$

the Taylor series for f about $x = x_0$.

Example: Find the Taylor series of the function $f(x) = \cos x$ at $x_0 = \frac{\pi}{2}$ and also find P_0, P_1, P_2, P_3, P_4 polynomials.

Solution: Given, $f(x) = \cos x$ $\qquad f\left(\frac{\pi}{2}\right) = \cos\frac{\pi}{2} = 0$

$$
\begin{aligned}
f'(x) &= -\sin x & f'\left(\frac{\pi}{2}\right) &= -\sin\frac{\pi}{2} = -1 \\
f''(x) &= -\cos x & f''\left(\frac{\pi}{2}\right) &= -\cos\frac{\pi}{2} = 0 \\
f'''(x) &= \sin x & f'''\left(\frac{\pi}{2}\right) &= \sin\frac{\pi}{2} = 1 \\
f^{iv}(x) &= \cos x & f^{iv}\left(\frac{\pi}{2}\right) &= \cos\frac{\pi}{2} = 0
\end{aligned}
$$

Now, the polynomials:

$$
\begin{aligned}
P_0(x) &= f(x_0) = f\left(\frac{\pi}{2}\right) = 0 \\
P_1(x) &= f(x_0) + f'(x_0)(x - x_0) \\
&= f\left(\frac{\pi}{2}\right) + f'\left(\frac{\pi}{2}\right)\left(x - \frac{\pi}{2}\right) \\
&= 0 + (-1)\left(x - \frac{\pi}{2}\right) \\
&= -\left(x - \frac{\pi}{2}\right) \\
P_2(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 \\
&= f\left(\frac{\pi}{2}\right) + f'\left(\frac{\pi}{2}\right)\left(x - \frac{\pi}{2}\right) + \frac{f''\left(\frac{\pi}{2}\right)}{2!}\left(x - \frac{\pi}{2}\right)^2 \\
&= 0 + (-1)\left(x - \frac{\pi}{2}\right) + \frac{0}{2!}\left(x - \frac{\pi}{2}\right)^2 \\
&= -\left(x - \frac{\pi}{2}\right) \\
P_3(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 \\
&= f\left(\frac{\pi}{2}\right) + f'\left(\frac{\pi}{2}\right)\left(x - \frac{\pi}{2}\right) + \frac{f''\left(\frac{\pi}{2}\right)}{2!}\left(x - \frac{\pi}{2}\right)^2 + \frac{f'''\left(\frac{\pi}{2}\right)}{3!}\left(x - \frac{\pi}{2}\right)^3 \\
&= 0 + (-1)\left(x - \frac{\pi}{2}\right) + \frac{0}{2!}\left(x - \frac{\pi}{2}\right)^2 + \frac{1}{3!}\left(x - \frac{\pi}{2}\right)^3 \\
&= -\left(x - \frac{\pi}{2}\right) + \frac{1}{3!}\left(x - \frac{\pi}{2}\right)^3 \\
P_4(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \frac{f^{iv}(x_0)}{4!}(x - x_0)^4 \\
&= f\left(\frac{\pi}{2}\right) + f'\left(\frac{\pi}{2}\right)\left(x - \frac{\pi}{2}\right) + \frac{f''\left(\frac{\pi}{2}\right)}{2!}\left(x - \frac{\pi}{2}\right)^2 + \frac{f'''\left(\frac{\pi}{2}\right)}{3!}\left(x - \frac{\pi}{2}\right)^3 + \frac{f^{iv}\left(\frac{\pi}{2}\right)}{4!}\left(x - \frac{\pi}{2}\right)^4 \\
&= 0 + (-1)\left(x - \frac{\pi}{2}\right) + \frac{0}{2!}\left(x - \frac{\pi}{2}\right)^2 + \frac{1}{3!}\left(x - \frac{\pi}{2}\right)^3 + \frac{0}{4!}\left(x - \frac{\pi}{2}\right)^4 \\
&= -\left(x - \frac{\pi}{2}\right) + \frac{1}{3!}\left(x - \frac{\pi}{2}\right)^3
\end{aligned}
$$

Taylor Series:

$$\sum_{k=0}^{\infty}\frac{f^{(k)}\left(\frac{\pi}{2}\right)}{k!}\left(x-\frac{\pi}{2}\right)^{k} = f\left(\frac{\pi}{2}\right)+f'\left(\frac{\pi}{2}\right)\left(x-\frac{\pi}{2}\right)+\frac{f''\left(\frac{\pi}{2}\right)}{2!}\left(x-\frac{\pi}{2}\right)^{2}+\frac{f'''\left(\frac{\pi}{2}\right)}{3!}\left(x-\frac{\pi}{2}\right)^{3}+\frac{f^{iv}\left(\frac{\pi}{2}\right)}{4!}\left(x-\frac{\pi}{2}\right)^{4}+\cdots$$

$$= 0-\left(x-\frac{\pi}{2}\right)+\frac{0}{2!}\left(x-\frac{\pi}{2}\right)^{2}+\frac{1}{3!}\left(x-\frac{\pi}{2}\right)^{3}+\frac{0}{4!}\left(x-\frac{\pi}{2}\right)^{4}+\cdots$$

$$= -\left(x-\frac{\pi}{2}\right)+\frac{\left(x-\frac{\pi}{2}\right)^{3}}{3!}+\cdots$$

Example: Expand $y = e^{ax}$ in powers of $(x-1)$. [Taylor series]

Solution: Given, $y = e^{ax}$.

In powers of $(x-x_0)$ means Taylor series.

In powers of x means Maclaurin series.

Here, $x_0 = 1$. So, $f(x) = e^{ax}$, $\quad f(1) = e^{a(1)} = e^{a}$

$$\begin{aligned} f'(x) &= ae^{ax}, & f'(1) &= a\times e^{a(1)} = ae^{a} \\ f''(x) &= a^2e^{ax}, & f''(1) &= a^2\times e^{a(1)} = a^2e^{a} \\ f'''(x) &= a^3e^{ax}, & f'''(1) &= a^3\times e^{a(1)} = a^3e^{a} \end{aligned}$$

$\vdots$

$$f^{(k)}(x) = a^ke^{ax}, \quad f^{(k)}(1) = a^k\times e^{a(1)} = a^ke^{a}$$

So, the Taylor series expansion of $y = e^{ax}$ in powers of $(x-1)$,

$$\sum_{k=0}^{\infty}\frac{f^{(k)}(1)}{k!}(x-1)^k$$

$$= f(1)+f'(1)(x-1)+\frac{f''(1)}{2!}(x-1)^2+\frac{f'''(1)}{3!}(x-1)^3+\cdots+\frac{f^{k}(1)}{k!}(x-1)^k+\cdots$$

$$= e^{a}+ae^{a}(x-1)+\frac{a^2e^{a}}{2!}(x-1)^2+\frac{a^3e^{a}}{3!}(x-1)^3+\cdots+\frac{a^ke^{a}}{k!}(x-1)^k+\cdots$$

H.W. 1. Expand the function $f(x) = 2x^3+7x^2-x-1$ in powers of $(x-2)$.

Answer: $41+51(x-2)+19(x-2)^2+2(x-2)^3+\cdots$

2. Expand the function $f(x) = sinx$ in powers of $\left(x-\frac{\pi}{2}\right)$ in infinite Taylor series.

Answer: $1-\frac{\left(x-\frac{\pi}{2}\right)^2}{2!}+\frac{\left(x-\frac{\pi}{2}\right)^4}{4!}-\cdots$

3. Expand $\frac{1}{x+2}$ at $x_0 = 4$.

Answer: $\frac{1}{6}-\frac{x-4}{36}+\frac{(x-4)^2}{216}-\frac{(x-4)^3}{1296}+\cdots$

H.W. Practice Sheet

H.W. Exercise 9.7 (17-24) and Exercise 9.8 (11-18)

Maclaurin Polynomial: (A special case of Taylor Polynomial)

If f can be differentiated n times at 0, then we define the nth Maclaurin polynomial for f to be

$$\begin{aligned} P_n(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots + \frac{f^{(n)}(0)}{n!}x^n \\ &= \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!}x^k \quad \text{[Summation Form]} \end{aligned}$$

Here, $x_0 = 0$.

Here,

$$\begin{aligned} P_0(x) &= f(0), \\ P_1(x) &= f(0) + f'(0)x, \\ P_2(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2, \\ P_3(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3. \end{aligned}$$

$\vdots$

Maclaurin Series:

In the special case where $x_0 = 0$, the Taylor series becomes,

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}x^k = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(k)}(0)}{k!}x^k + \cdots$$

in which case we call it the Maclaurin series for f.

Example: Find the Maclaurin polynomials P_0, P_1, P_2, P_3 for $y = e^x \cos x$.

Solution: Given, $y = f(x) = e^x \cos x$ $\qquad f(0) = e^0 \cos 0 = 1 \cdot 1 = 1$

$$\begin{aligned} f'(x) &= e^x \cos x - e^x \sin x = e^x(\cos x - \sin x), \quad f'(0) = e^0(\cos 0 - \sin 0) = 1(1-0) = 1 \\ f''(x) &= e^x(\cos x - \sin x) + e^x(-\sin x - \cos x) \\ &= e^x(\cos x - \sin x - \sin x - \cos x) \\ &= -2e^x \sin x \end{aligned}$$

$f''(0) = -2e^0 \sin 0 = 0$

$$f'''(x) = -2e^x \sin x - 2e^x \cos x$$
$$f'''(x) = -2e^x(\sin x + \cos x)$$

$$f'''(0) = -2e^0(\sin 0 + \cos 0)$$
$$= -2 \times 1 \times (0 + 1)$$
$$= -2$$

Now, the polynomials,

$$P_0(x) = f(x_0) = f(0) = 1$$
$$P_1(x) = f(0) + f'(0)x = 1 + 1 \cdot x = 1 + x$$
$$P_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2$$
$$= 1 + 1 \cdot x + \frac{0}{2!}x^2$$
$$= 1 + x$$
$$P_3(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3$$
$$= 1 + 1 \cdot x + \frac{0}{2!}x^2 + \frac{-2}{3!}x^3$$
$$= 1 + x - \frac{2}{6}x^3$$
$$= 1 + x - \frac{x^3}{3}$$

Example: Expand $y = \ln(x + 1)$ in powers of x. [Maclaurin Series]

OR Find the Maclaurin series for $y = \ln(x + 1)$ at $x_0 = 0$.

OR Find the Maclaurin series for $y = \ln(x + 1)$.

In powers of $(x - x_0)$ means Taylor series.

In powers of x means Maclaurin series.

Solution: Given,

$$y = f(x) = \ln(x + 1), \qquad f(0) = \ln(0 + 1) = \ln(1) = 0$$
$$f'(x) = \frac{1}{x + 1} = (x + 1)^{-1}, \quad f'(0) = \frac{1}{0 + 1} = 1$$
$$f''(x) = -(x + 1)^{-2}, \quad f''(0) = -(0 + 1)^{-2} = \frac{-1}{1^2} = -1$$
$$f'''(x) = 2(x + 1)^{-3}, \quad f'''(0) = 2(0 + 1)^{-3} = \frac{2}{1^3} = 2$$

$\vdots$

The Maclaurin Series:

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

$$= 0 + 1 \cdot x - \frac{1}{2!}x^2 + \frac{2}{3!}x^3 + \cdots$$

$$= x - \frac{x^2}{2} + \frac{2}{6}x^3 - \cdots$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

H.W. 1. Find the Maclaurin series for $\frac{1}{1-2x}$.

Answer: $1 + 2x + 4x^2 + 8x^3 + \cdots$

2. Expand the function $f(x) = \ln\left(\frac{1+x}{1-x}\right)$ infinite Maclaurin series.

Answer: $2x + \frac{2x^3}{3} + \frac{2x^5}{5} + \frac{2x^7}{7} + \cdots$

3. Expand the function $f(x) = e^{cosx}$ at $x_0 = 0$.

Answer: $e - \frac{ex^2}{2} + \frac{ex^4}{6} - \frac{31ex^6}{720} + \cdots$

H.W. Practice Sheet

H.W. Exercise 9.7 (7-12) and Exercise 9.8 (1-6)